

Progress Test 1.B — Thermo-S1

1.B · Postulates of thermodynamics

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Temperature conversion

(1 mark)

Liquid nitrogen boils at $T = 77\text{ K}$ under atmospheric pressure. Express this temperature in degrees Celsius, exactly.

$\theta = T - 273.15 = 77 - 273.15$	$\theta = -196.15\text{ }^{\circ}\text{C}$
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Marking

①
0 1

Marking scheme

① 1 pt $-196.15\text{ }^{\circ}\text{C}$ (the minus sign is part of the answer).

Examiner's tip

The kelvin is written with a capital K and no degree sign: “77 K”, not “77 °K”. Test 1's marker noted: “big K stands for kelvins”.

Exercise 2. Thermodynamic equilibrium — postulates

(5 marks)

A system is in contact with surroundings at pressure p_0 and temperature T_0 . Give the two conditions of thermodynamic equilibrium, with the corresponding equalities. State the postulates of classical thermodynamics. When are the state variables of a system defined?

Thermal equilibrium: $T_{\text{system}} = T_0$. Mechanical equilibrium: $p_{\text{system}} = p_0$ (for a movable wall). Postulates. (1) A system left to itself (isolated, or in constant surroundings) evolves spontaneously towards an equilibrium state, and then no longer leaves it. (2) At equilibrium, the state of the system is completely described by a <i>small number</i> of macroscopic state variables (p, V, T, n). There is no need for the positions and velocities of all the particles. The state variables are defined (uniform in space and constant in time) only at equilibrium. Between two equilibrium states they may not be defined.
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Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① **2 pts** $T = T_0$ (thermal) and $p = p_0$ (mechanical), 1 pt each.
- ② **2 pts** Evolution towards equilibrium (1 pt); a few state variables suffice (1 pt).
- ③ **1 pt** State variables are defined only at equilibrium.

Note

Box 8 is blank in the handout. These are the standard statements; accept the wording given in class.

Exercise 3. Equilibrium or not?

(4 marks)

A syringe with a free, sealed piston is left for a long time in a room at (p_0, T_0) . The enclosed gas then has:

- | | |
|---|--|
| ☒ $p = p_0$ and $T = T_0$ | ☐ $p > p_0$ and $T = T_0$ |
| ☐ $p = p_0$ and $T > T_0$ | ☐ p and T not defined |

Why: once equilibrium is reached, the state variables of the system equal those of the surroundings (Fig. 3 and remark (i)).

Just after a sudden push on the piston, the pressure of the gas:

- | | |
|--|---|
| ☐ already has its final value | ☒ is not defined: it is not uniform |
| ☐ is equal to p_0 | ☐ is zero |

Why: it takes some time for p and T to become uniform and steady again: state variables exist only at equilibrium.

A metal rod has one end in boiling water and the other in melting ice. Its temperature profile no longer changes. The rod is:

- | | |
|---|---|
| ☐ at thermodynamic equilibrium | ☒ in a steady state, not at equilibrium |
| ☐ isolated | ☐ at equilibrium: nothing changes |

Why: equilibrium requires both a steady state and homogeneity. Here T varies along the rod.

To describe a closed system at equilibrium, classical thermodynamics uses:

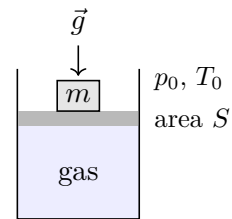
- | | |
|---|---|
| ☐ all positions and velocities | ☒ a few state variables (p, V, T) |
| ☐ only its mass | ☐ its whole past history |

Why: this is the second postulate (remark (ii) of the notes).

Exercise 4. A loaded piston

(4 marks)

A vertical cylinder is closed by a frictionless piston of area S and negligible mass. The surroundings are at p_0 and T_0 . A mass m is placed on the piston. Find the pressure and the temperature of the gas once equilibrium is reached.



$$p_0 = 1.0 \times 10^5 \text{ Pa} \quad T_0 = 293 \text{ K} \quad m = 1.0 \text{ kg}$$

$$S = 2.0 \text{ cm}^2 \quad g = 10 \text{ m s}^{-2}$$

Equilibrium of the piston: $pS = p_0S + mg \Rightarrow p = p_0 + \frac{mg}{S}$

$$S = 2.0 \text{ cm}^2 = 2.0 \times (10^{-2} \text{ m})^2 = 2.0 \times 10^{-4} \text{ m}^2$$

$$\frac{mg}{S} = \frac{1.0 \times 10}{2.0 \times 10^{-4}} \text{ Pa} = 5.0 \times 10^4 \text{ Pa}$$

$$p = 1.5 \times 10^5 \text{ Pa}$$

and, once thermal equilibrium is reached,

$$T = T_0 = 293 \text{ K}$$

Marking

①		
0	1	
②		
0	1	
③		
0	1	
④		
0	1	

Marking scheme

- ① 1 pt Force balance on the piston: $p = p_0 + mg/S$ (literal).
- ② 1 pt $S = 2.0 \times 10^{-4} \text{ m}^2$.
- ③ 1 pt $p = 1.5 \times 10^5 \text{ Pa}$.
- ④ 1 pt $T = T_0 = 293 \text{ K}$ (thermal equilibrium).

Examiner's tip

$1 \text{ cm}^2 = (10^{-2} \text{ m})^2 = 10^{-4} \text{ m}^2$, not 10^{-2} . As in Test 1: keep letters until the end, then substitute with units.

Your score: / 14 $\rightarrow$ mark out of 20 = score $\times$ 20/14 = / 20